

Swaption model		1	into	3		
Time	Swap rate	zc	Period	$DF(0, t)$		
1	5.000%	5.000%	0x1		d_1	0.1000
2	5.110%	5.113%	1x2	0.9051	d_2	-0.1000
3	5.300%	5.312%	2x3	0.8562		
4	5.470%	5.494%	3x4	0.8074		
F	5.644%		$A =$	2.5687		
σ	20.0%				$XN(-d_2) - FN(-d_1)$	0.004496
T	1		Notional	100,000,000	Receiver swaption	1,154,863
X	5.644%				(in bps)	115.5

Figure 5.3: Pricing Up the 1 into 3 Receiver Swaption

5.644%, as demonstrated in Figure 5.3. Note that since both swaptions are struck at-the-money forward, they have the same value as we know must be the case from swaption parity. $\square$

5.2 The Link Between Caps/Floors and Swaptions

Suppose that an investor is interested in buying an option to protect against rates going higher. Both caps and payer swaptions come to mind, but which one is more appropriate depends on the needs of the investor.

The buyer of a cap is buying a series of options. For example, the buyer of a 5-year cap on 3-month LIBOR is buying a series of 20 different options. Each option in the series provides protection against a move up in a distinct setting of 3-month LIBOR. Whereas one LIBOR setting in the series may be low, resulting in an out-of-the-money option one period, that does not affect the ability of another setting to be above the strike, resulting in an in-the-money payout in another period.

The buyer of a payer swaption, on the other hand, is buying a single option. For example, a 5 into 10 payer swaption will protect the buyer from an increase in 10-year swap rates, five years from now. And whereas an in-the-money setting on a caplet accrues for three months as per Equation 4.1,

an in-the-money setting on a swaption will result in payments over the entire life of the underlying swap.¹⁶

The choice of which option is more appropriate depends on the needs of the particular investor. If, for example, the investor manages a bank portfolio that has a liability tied to 3-month LIBOR resetting quarterly over the next five years, then it would seem more appropriate for the bank to purchase a 5-year cap on 3-month LIBOR instead of purchasing a swaption. If, however, the investor manages a mortgage portfolio that has exposure to 10-year swap rates increasing, say, five years from now, then it would appear more appropriate for the investor to buy a 5 into 10 payer swaption rather than a cap.¹⁷ So whereas caps and payer swaptions both offer protection against higher rates, their use varies in accordance with the needs of investors.

There is an intimate relationship between caps and payer swaptions and also between floors and receiver swaptions. Consider now a 2x5 cap on 3-month LIBOR (a forward starting cap) and a 2 into 3 payer swaption (with the same strike, quoted quarterly money versus 3-month LIBOR). These two trades are represented in Figure 5.4.

The 2x5 cap has its first reset in two years and its final payment in five years. The trade is composed of 12 options, covering 12 3-month LIBOR settings as indicated in the top half of this figure. By comparison, the 2 into 3 payer swaption is a single option that expires in two years and is exercisable into a 3-year swap, as indicated in the bottom half of this figure. The underlying in this swaption is the 3-year swap rate, two years forward. If we think about how we price up forward starting swaps as we did in Section 3.3.4, the underlying 3-year swap, two years forward is determined by exactly the same 12 3-month LIBOR forwards as comprise the 2x5 cap.

So what's the difference between a 2x5 cap and a 2 into 3 payer swaption? We've already talked about the different ways in which the two trades are used by investors. From a pricing perspective, notice that the 2x5 cap must be worth at least as much as the 2 into 3 payer swaption.¹⁸ The cap is a

¹⁶If the payer swaption is physically settled, then upon exercise the owner of the option will pay a fixed rate equal to the strike and receive LIBOR over the life of the swap. If instead the swaption is cash settled, then instead of actually entering into this swap, the owner of the option is paid the mark-to-market value of this swap (which we know how to compute from Section 3.3.1). Although these two methods of exercise result in dramatically different cash flows, their value at the point of exercise is identical.

¹⁷Actually, in this scenario the portfolio most likely has exposure to an increase in 10-year swap rates over the next five years, not just exactly five years from now. An increase in swap rates over the life of the swaption will serve to increase its value, thereby buffering any ill effects the increase has on the value of the mortgage portfolio itself.

¹⁸To show this is true, assume that the cap costs less than the payer. Then we could buy the cap and sell the payer, taking in money today. Let's consider two cases. First, if in two years the payer is not exercised against us, then we will have nonnegative cash

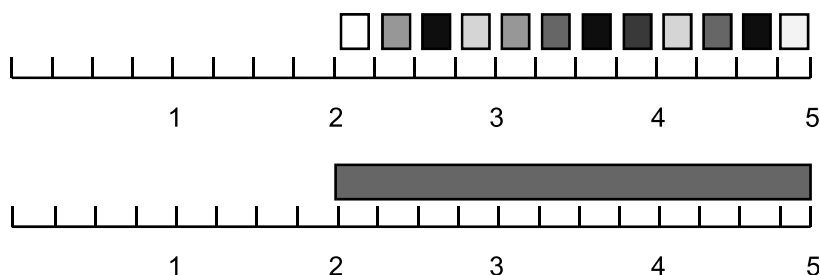


Figure 5.4: Representation of 2x5 Cap (*top diagram*) and a 2 into 3 Payer Swaption (*bottom diagram*)

portfolio of options, and the payer swaption is an option on a portfolio of forward rates. That is, the 2x5 cap is a portfolio of options on 12 different rates. In comparison, the 2 into 3 payer swaption is an option on a portfolio of forward rates (i.e., the 12 3-month LIBOR forwards that comprise the formulation of the underlying 3-year swap, two years forward). Thus caps and payers are intimately linked. In fact, a popular relative value trade is to buy one of these options versus selling the other, thereby taking a view on the value of one of these options versus the other. When we get to Chapter 9, we will return to this example to discuss another difference between the two trades.

5.3 Questioning Black's Model for Interest Rate Options

As we have discussed, Black's model, a simple extension of Black-Scholes, has traditionally been a workhorse model for the pricing of a variety of interest rate options. But what if Black-Scholes is wrong?¹⁹ Sacrilege? Perhaps. But

flows in the future (and we may make some money as well if any of our caplets ever set in-the-money). Second, suppose that in two years the payer is exercised against us. Then for each of the next quarterly periods, we will be receiving fixed at the strike and paying 3-month LIBOR. That is, our cash flow from the payer swaption being exercised against us will be $\text{Strike} - \text{LIBOR}$. Of course, we are still long our cap, which will pay us $\max(\text{LIBOR} - \text{Strike}, 0)$ each quarter. Thus, our net cash flow from the cap and the payer is

$$\text{Strike} - \text{LIBOR} + \max(\text{LIBOR} - \text{Strike}, 0),$$

which is nonnegative. Thus, we always have nonnegative cash flows in the future, and this would be an arbitrage. So it must be that the cap is worth no less than the payer.

¹⁹Inasmuch as Black's model is a relatively straightforward extension of Black-Scholes, people in industry often use the two names interchangeably. Reference to either Black's

there is good reason to question whether or not Black-Scholes is a “good” model to employ for the pricing of interest rate options. Actually, we will discuss three such reasons: the appropriateness (or rather, inappropriateness) of the assumption that rates are lognormal, the (sometimes) surprising relationship between swaption prices and implied volatilities when computed using Black’s formula, and most important, the existence of skew.

With a significant amount of doubt cast over the appropriateness of Black’s model for the pricing of interest rate options, the question is, What model for the pricing of such options is more appropriate? We will introduce the concept of the normal model, in which rates are assumed to be normally distributed as an alternative.

5.3.1 Are Interest Rates Lognormal?

Did you ever notice that stocks that have a relatively high price tend to have greater daily price fluctuations than stocks that are relatively low in price? If you’ve ever looked at a stock with a really high price, say, in the \$400 range, you will notice that the daily price fluctuations in such a stock can be much larger than those that have low prices, say, in the \$10 range. A stock that is in the \$400 price range may have daily price fluctuations as high as \$10 or \$20, or even higher. Low price stocks, say, those in the \$10 range, don’t typically have that magnitude of daily price changes. Most \$10 stocks have daily price fluctuations that are a fraction of a dollar.

The phenomenon that we just referred to is characteristic of asset prices that follow a lognormal distribution.²⁰ In Appendix A, we discuss that if an asset price, S_T , is lognormally distributed, we can write

$$S_T = S e^{(\mu - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}z}, \quad (5.17)$$

where z is a standard normal random variable and S is today’s asset price. The standard deviation of the asset price is given by

$$S e^{\mu T} \sqrt{(e^{\sigma^2 T} - 1)}. \quad (5.18)$$

model or Black-Scholes implies a model in which the underlying forward rate is assumed to be lognormally distributed and the volatility of the underlying is constant and does not vary with strike. Black’s formula is the resultant equation to price options that follows from these assumptions.

²⁰Although the phenomenon that we just discussed is suggestive of the lognormality of stock prices, strictly speaking this is a bastardization of the term *lognormality* as it refers to the distribution of stock prices. To say that the distribution of a stock price is lognormal is a statement of the characteristics of an individual stock price in and of itself. It is not a statement comparing two different stocks. Technically, the potential lognormality of stock prices would be better demonstrated by looking at the distributional characteristics of a single stock.

From Equation 5.18, we see that the standard deviation of the asset price increases with the level of the asset price. That is, under the assumption of lognormality an asset price has a higher standard deviation, the greater the level of the asset price. When we are working with Black-Scholes, the volatility parameter σ is a percentage; it is a measure of *relative volatility*, meaning that the volatility of the underlying is relative to the level of the underlying.

The distribution of swap rates, however, does not typically exhibit this phenomenon. Figure 5.5 shows a graph of daily changes in 10-year swap rates. The panel on the top samples daily changes during a period of time when 10-year swap rates were relatively low, and the panel on the bottom shows a sample when rates were relatively high. The distribution of daily changes from one panel to the next does not look remarkably different from one another. This would tend to refute the notion that 10-year swap rates are lognormally distributed. If 10-year swap rates were lognormally distributed, we would tend to observe larger daily rate fluctuations in a high-rate environment than in a low-rate environment.²¹

An Alternative Distributional Assumption for Interest Rates: Normality

There is a well-known distribution, however, that exhibits the characteristic that volatility does not depend on the level of rates. Under a normal distribution, the volatility of swap rates has no dependency on the level of rates. At the risk of getting ahead of ourselves, if a swap rate R to be observed at time T is normally distributed, we can posit a model in which

$$R \sim N(\mu, \sigma_N^2 T), \quad (5.19)$$

where μ and $\sigma_N^2 T$ are the mean and variance of the distribution, respectively.²² The parameter σ_N is referred to by various names, including *normal vol* or *normalized vol* or *basis point vol* or *annualized basis point vol*. Note that in this formulation the standard deviation of the swap rate,

²¹We should caution that this analysis is the most casual of empirical studies. Our purpose here is not to provide rigorous statistical tests crediting or discrediting the assumption of lognormality for the distribution of swap rates. Rather, what we hope to accomplish here is to show why those in the industry are often comfortable with the assumption of lognormality for certain asset classes, such as stocks, but are much less comfortable making this distributional assumption for interest rates.

²²A random variable R is said to be normally distributed with mean μ and variance σ^2 , denoted $R \sim N(\mu, \sigma^2)$, if it has the probability density function

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}.$$

See Ross (2009) for more details.

SWAPTIONS

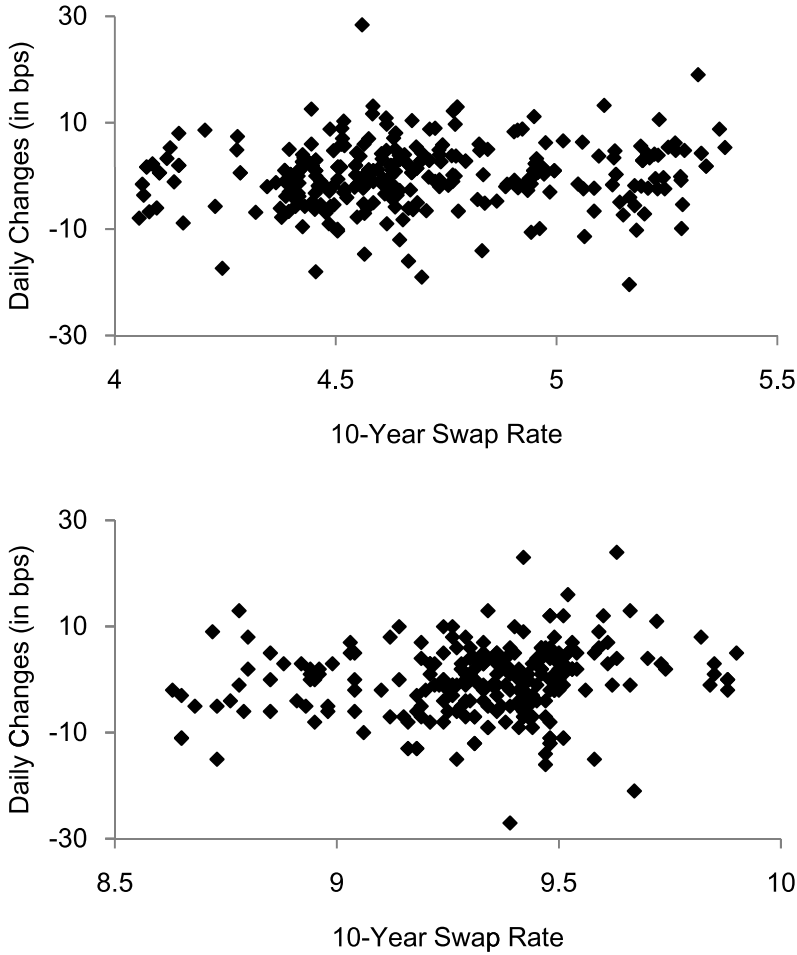


Figure 5.5: Daily Changes in 10-Year Swap Rates in Both a Low-Rate (*top half*) and a High-Rate (*bottom half*) Environment

$\sigma_N \sqrt{T}$, does not depend on the level of the swap rate. A model for pricing interest rate options in which rates are assumed to be normally distributed is referred to as the *normal model*. The implied volatility parameter σ_N obtained from pricing options using the normal model is referred to as *normal implied volatility*. We will look at this model extensively in Section 5.4.

Under the normal model, basis point volatility, σ_N , is assumed to be a constant. In the normal model, σ_N is a measure of *absolute volatility*. Whereas the parameter σ in Black's model is usually expressed as a percent-

age (it is a measure of *relative volatility*), it is often useful to think of σ_N in the normal model in terms of basis points. The empirical results depicted in Figure 5.5 (in which the daily fluctuations in swap rates do not depend on the level of rates) appear to be more consistent with a normal model than a lognormal model.

5.3.2 Swaption Prices and Implied Vol

A second unsettling feature of a lognormal model for the pricing of options on rates can be seen in Figure 5.6. This figure shows a history of at-the-money forward 1 into 10 swaption straddle prices and the associated history of implied vols under two different models. The top panel shows the history of *lognormal implied vols*. That is, given the swaption straddle prices observed in the market, and under the assumption that swap rates are lognormal, implied volatilities are computed using Black's model as in Equations 5.9 and 5.10. A disturbing feature of this picture is that there are times when lognormal implied vols and swaption prices move inversely. For example, near the very end of the time series, we see that although swaption straddle prices were going up, lognormal implied vols are going down. This runs counter to our usual intuition that option prices and implied vols move in the same direction and is suggestive that maybe a lognormal model for interest rate options does not explain everything.

By comparison, the bottom panel shows the same history of swaption prices in conjunction with a history of normal implied vols. These normal implied vols are computed from an option pricing model in which it is assumed that swap rates are normally distributed as given by Equation 5.19.²³ As this figure demonstrates, normal implied vols tend to move in the same direction as swaption prices, which is intuitively pleasing and suggestive that perhaps a normal model explains things that the lognormal model does not.

5.3.3 Skew

Suppose that we were to gather market prices on a bunch of different swaptions on the same underlying swap rate and time to maturity, allowing only strikes to differ. Suppose further that we compute the implied volatility for each of these swaptions, and graph the implied volatility as a function of the strike price.²⁴ Under the assumptions of Black's model, this graph should be

²³We will develop such an option pricing model, known as the normal model, in Section 5.4.

²⁴Once again, given the price of an option, the *implied volatility* is the volatility that we plug into a pricing model to obtain the price of the option.

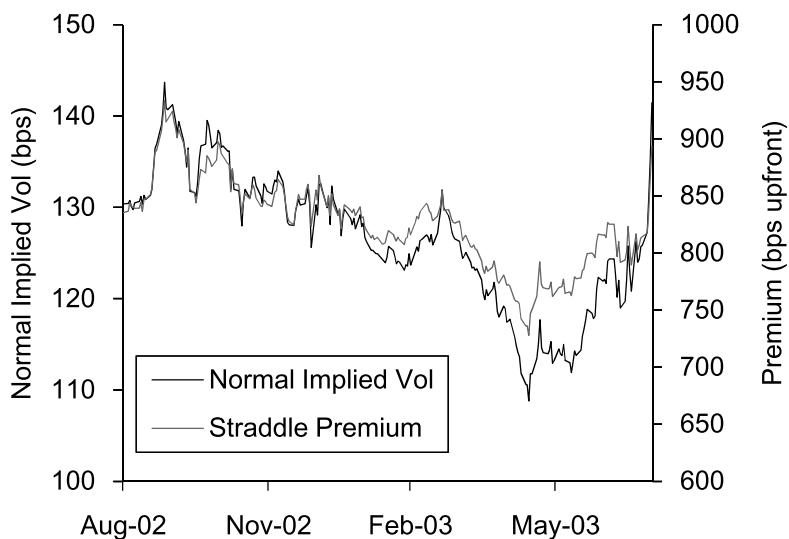
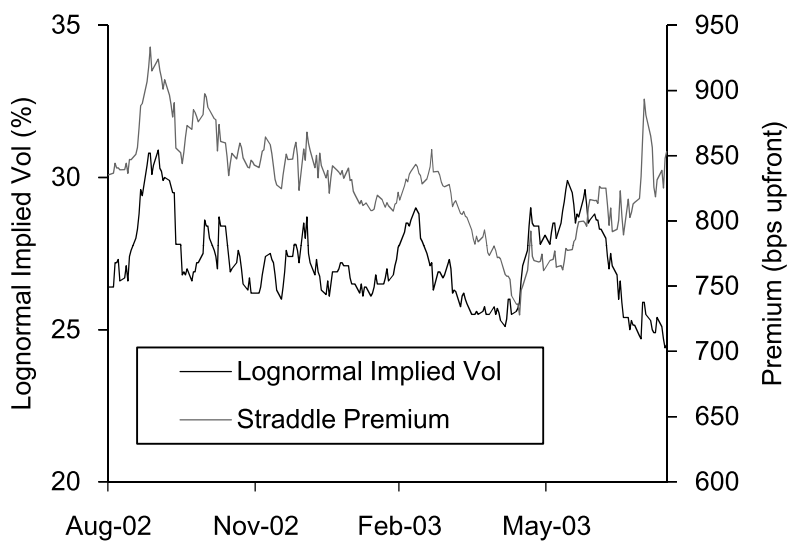


Figure 5.6: The Relationship Between Swaption Prices and Lognormal Implied Vol (*top half*) and Normal Implied Vol (*bottom half*)

a flat line. That is, Black's model says that the lognormal implied volatility of an option should be a constant that does not vary with the level of the strike.

But in reality this is often not the case. The pattern that is most often observed in the swaption market is one in which low-strike options trade at a lognormal implied vol above that of the at-the-money strike, and high-strike options trade at a lognormal implied vol below that of the at-the-money strike. *Skew* refers to the relationship between the implied vol of an option and its strike as we allow the strike to vary. The pattern just described is referred to generically as *downward skew*. Figure 5.7 demonstrates such a pattern observed from computing lognormal implied vols from swaption prices trading in the market. This figure is obtained by observing actual prices that traded in the market, then using Black's formula to compute lognormal implied volatilities. A striking (no pun intended) feature of this figure is that these lognormal implied vols exhibit downward skew. Yet Black's model

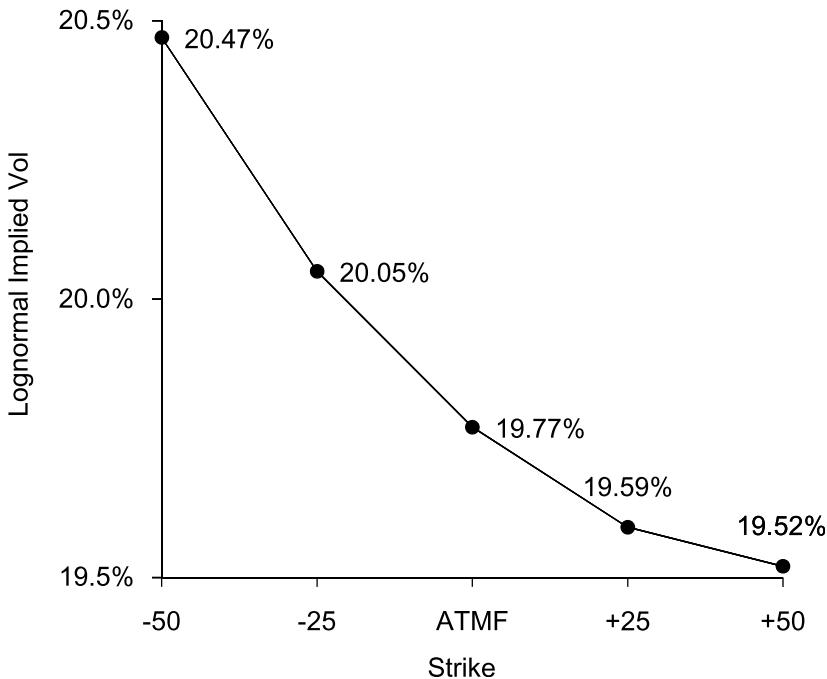


Figure 5.7: Lognormal Implied Vol as a Function of Strike for 6 Month into 5-Year Swaptions on November 18, 2005

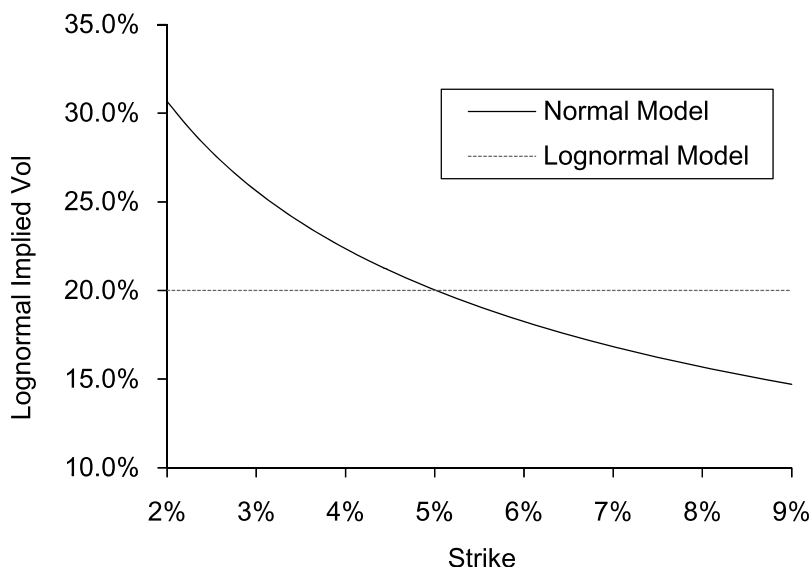


Figure 5.8: Flat Skew of the Black (Lognormal) Model and Downward Skew of the Normal Model

makes the assumption of *flat skew*, meaning it is assumed that there is no variation in implied vol for different strikes.

The normal model, on the other hand, implies a skew pattern in which low-strike options have lognormal implied vols (also called Black implied vols) above that of the at-the-money strike, and high-strike options have lognormal implied vols below that of the at-the-money strike.²⁵ The skew patterns of these competing theoretical models are demonstrated in Figure 5.8. If Black's model is true, then of course lognormal implied vol is a constant that does not vary with strike. This is depicted by the flat line in Figure 5.8. If, on the other hand, we assume that the normal model is true (i.e., we assume that swap rates are normally distributed and that σ_N is constant as we allow the strike to vary), the implication is that lognormal implied vol from Black's model must exhibit downward skew. This of course is depicted by the sloped line in Figure 5.8. Notice that the theoretical pattern of lognormal implied vols under the normal model in Figure 5.8 closely resembles the actual pattern sometimes observed in the market as depicted in Figure 5.7.

²⁵We will examine the normal model closely and derive this relationship ourselves in Section 5.4.5.